

BDRRMO Interview — Summary, Interpretation, and Key Findings

Interviewee: Johnny P., BDRRMO Staff

Date: May 12, 2026 | 12:54 PM – 2:00 PM

Location: City of Bacoor Legislative and Disaster Resilience Building, Brgy. Bayanan,
Bacoor City

I. SUMMARY

The interview was conducted with a BDRRMO staff member as part of the thesis team's problem identification and validation phase for their proposed mobile application, *FloodSense*. Rather than functioning as a straightforward question-and-answer session, the interview evolved into a rigorous technical discussion where the interviewee simultaneously provided organizational insights and critically evaluated the research team's conceptual understanding of their own project.

The interviewee described BDRRMO's role as primarily **monitoring, coordination, and dissemination** — tracking incoming weather disturbances, coordinating with barangays, and cascading flood warnings through social media, email, and direct barangay communication. The office follows established protocols, contingency plans, and a city ordinance (City Ordinance 246-2022) that enables preemptive evacuation before flooding occurs. For flood monitoring, BDRRMO uses **GIS-based tools**, Climate and Disaster Risk Assessment (**CDRA**) **field assessments**, and manual rainfall gauging in Niog. Data is stored in a database system that integrates Excel and GIS formats. Available data that could be shared with the research team includes **barangay boundary maps, historical flood location records, and exposure maps** — but no real-time or sensor-based data exists.

On the proposed FloodSense application, the interviewee raised numerous conceptual and technical concerns — particularly around user input accuracy, the AR mechanism, output definition, sustainability, and the distinction between flood susceptibility

and flood risk. Despite his criticisms, he consistently guided the researchers toward workable solutions and implicitly affirmed that the project addresses a genuine community need. He ended the interview indicating openness to collaboration and data sharing.

II. KEY FINDINGS PER INTERVIEW QUESTION

Q1 — Office Role in Flood Monitoring and Disaster Response BDRRMO's role covers the full disaster management cycle: pre-disaster monitoring and contingency planning, response coordination during flooding, and post-event assessment. They monitor threats through social media, emails, and inter-agency networks, then disseminate warnings downward to barangays. City Ordinance 246-2022 authorizes preemptive evacuation, which the interviewee cited as an effective policy mechanism for zero-casualty targeting.

Q2 — Flood-Related Data Collected The office collects rainfall data (manual gauge in Niog), flood level markers, and records of affected areas during flood events. These are primarily historical and observational in nature, not real-time or sensor-based.

Q3 — Data Storage Format Data is stored in a database system that incorporates both Excel spreadsheets and GIS layers. The interviewee personally specializes in GIS and cited it as the office's primary spatial analysis tool.

Q4 — Tools and Systems for Flood Mapping or Risk Assessment BDRRMO uses GIS for spatial flood mapping and the CDRA (Climate and Disaster Risk Assessment) framework for field-based risk evaluation. These are the core tools used in identifying flood-prone zones and planning disaster response strategies.

Q5 — Consistently Flood-Prone Barangays District 1 barangays — particularly those in the Zapote and Talaba areas — were identified as consistently flood-prone. The interviewee emphasized that **low elevation** is the primary physical driver of flood susceptibility in Bacoar City. He explained that areas at lower elevations accumulate water from surrounding

higher-elevation zones, meaning flooding can occur even with minimal local rainfall if upstream or upland areas receive heavy precipitation.

Q6–Q7 — Warning Communication Methods and Effectiveness BDRRMO disseminates warnings through social media, email, and direct coordination with barangay officials, who then use PA systems and word-of-mouth. The interviewee rated these methods as **very effective** for reaching the population they serve, citing their historical data as a strong basis for strategic positioning of resources before disasters.

Q8 — Biggest Challenges Residents Face The primary behavioral challenge is **residents refusing to evacuate** — often due to fear of theft or reluctance to leave elderly family members behind. The interviewee also noted that the city's preemptive evacuation ordinance has helped address this. He acknowledged that commuters and people outside their home areas remain a vulnerable group that a mobile app could meaningfully serve.

Q9 — Limitations in Collecting, Analyzing, or Sharing Flood Information Key limitations include: difficulty gathering community participation (residents avoiding interviews or orientations), poverty-related barriers (residents prioritizing livelihood over attending flood briefings), and limitations in real-time weather data capture.

Q10 — Biggest Gap in Flood Information Delivery The interviewee identified **community engagement and information reach** as the core gap — residents who most need awareness are often the hardest to reach through formal channels. He noted that those who do attend orientations are sometimes not genuinely engaged.

V1 — Validation of Survey Findings The interviewee accepted the survey's general findings but raised a **methodological concern** about flood depth measurement. He questioned the validity of "knee-deep" and "waist-deep" as flood depth categories, noting that these measurements vary significantly by individual height. He suggested that future data collection use standardized measurements such as inches or feet to ensure comparability. He described the current survey results as "tansya-meter" —

estimation-based — which, while not invalidating the findings, is a panel-level concern the research team should be prepared to address.

V2 — Genuine Need for a Mobile-Based Flood Tool The interviewee affirmed that a genuine need exists, particularly for **commuters, travelers, and residents unfamiliar with specific flood-prone zones across Bacoar**. He reframed the app's value proposition clearly: it is not just for people who are already flooded, but for anyone navigating the city who needs to make route decisions based on flood susceptibility. He also noted the app's potential for community-based data contribution, where local inputs about flooding conditions can inform other users in real time.

V3 — Openness to Research Collaboration The interviewee confirmed that BDRRMO can provide the following data to support the research:

Barangay boundary maps

Historical flood location records

Exposure maps (per barangay)

He was clear that the research team would need to submit a formal data request. He also stressed that the comparison between **projected susceptibility (from the ML model) and actual historical flood data (from BDRRMO)** would be the most meaningful output of the study — showing the difference between what the model predicts and what has actually occurred on the ground.

V4 — Endorsement as a Supplementary Flood Awareness Tool While not explicitly stated as a formal endorsement, the interviewee's overall stance was **conditionally supportive**. He implied that if the research team addresses the conceptual and methodological gaps he identified, the app could serve as a legitimate supplementary tool that BDRRMO could recommend — particularly for community awareness and commuter safety.

III. CRITICAL OBSERVATIONS AND INTERPRETATIONS

1. The interviewee functioned as an informal panel reviewer. His questioning style — pushing the researchers to defend their title, clarify their output, and explain their user input mechanism — closely mirrors what a thesis panel would do. This means the transcript is not only a data source but also a **rehearsal of panel objections** the team must be ready to address.

2. The AR mechanism needs better framing. The interviewee was initially confused about how augmented reality works in the context of the app. This signals that the research team must invest in clearer documentation and demonstration materials — both for panel defense and for user onboarding within the app itself.

3. User input accuracy is the app's most critical vulnerability. The interviewee consistently challenged how ordinary residents could accurately input rainfall data in millimeters. His proposed solution — **video-based rainfall intensity selection** where the app computes the corresponding mm value internally — is a practical, user-centered design recommendation that directly addresses this gap. This should be seriously considered in the system design.

4. Flood susceptibility and flood risk are not the same. The interviewee drew a clear distinction: susceptibility is physical (how prone is the land), while risk incorporates human vulnerability and community capacity. He suggested integrating a community capacity score derived from survey feedback to compute true risk levels. This is an academically significant distinction that the research team must reflect in their Chapter 2 conceptual framework.

5. Data sourcing responsibility must shift to the researchers. For soil type, drainage proximity, and elevation — the interviewee explicitly stated that these should be determined by the researchers using existing maps and GIS data, not sourced from community respondents. This simplifies the ML model's data pipeline and reduces survey burden.

6. The STANDL map standard is a required output framework. The interviewee introduced STANDL (Scale, Title, Author, North Arrow, Date, Legend) as the minimum requirement for any map output. The FloodSense map display must incorporate all these elements to be considered a credible cartographic output in the eyes of a technical evaluator or panel.

7. Outcome, not just output, must be defined. The interviewee repeatedly pressed the team on the *outcome* of the app — its long-term value beyond just rendering a map. The research team should articulate a clear outcome statement: e.g., *"To reduce flood-related wrong decisions among Bacoar City residents and commuters by providing accessible, location-specific flood susceptibility information."*

IV. SUMMARY OF ACTIONABLE RECOMMENDATIONS FROM THE INTERVIEWEE

Area	Recommendation
User input	Replace free-text/dropdown mm input with video-based rainfall intensity selection
AR surface detection	Add green/red surface detection indicator before allowing AR render
Map output	Apply STANDL standard; define map type (susceptibility map, zone-based)
ML data sourcing	Researchers self-source elevation, soil type, drainage data via GIS/maps
BDRRMO data	Submit formal request for barangay boundaries, historical flood records, exposure maps

Risk vs. susceptibility Define both terms separately; incorporate community capacity into risk computation

Outcome statement Clearly define long-term impact beyond the map output

Panel preparation Be ready to defend: title, output type, input accuracy, AR mechanism, sustainability

This summary is based on the verbatim transcript of the BDRRMO interview conducted on May 12, 2026, as part of the thesis study on FloodSense: An Augmented Reality Mobile Application with Machine Learning-Based Flood Susceptibility Mapping for Bacoar City.